Supplementary Material
PaTH-VLA: Progress-and-Target Context for Long-Horizon Robot Policy Adaptation

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A. Pre-registration

The following text was frozen and hashed (SHA-256, to be released with the code) before the first optimizer update and before the first scored episode. It is reproduced verbatim except for the removal of machine paths and, in this public version, the omission of one administrative scheduling line; the hash refers to the original file. It was written by the AI agent that executed the study (see the LLM Usage Statement of the main paper); "user" refers to the authors.

Pre-registration: Stage / Box+Location / Full adaptation of frozen RoboCasa365 policies

Frozen on 2026-11-08 (UTC) before any training run and before any scored evaluation episode of this study. Executor: Claude (Fable 5.1) on behalf of Qinzhen Ma; cluster hkust-cluster (8x H100 PCIe, GPUs 2/3 faulted and unused).

Question Does lightweight auxiliary supervision about *where the policy is in the task* (stage prediction) and *what it should attend to* (bounding-box prompts + target-location prediction) improve a pretrained generalist manipulation policy on long-horizon RoboCasa365 tasks, and which component carries the effect?

Models (official RoboCasa365 checkpoints, pinned in the weights index)

- Xiaomi Robotics 1 (XR1)
- Awomo-0.5I
- Awomo-0.5V (Awomo-I/V are related variants: two model families, three models)

Tasks (10, fixed; selected 2026-11-05 from published per-task success in [20%, 60%] for all three models)

- composite-seen (training demos available): DeliverStraw, GetToastedBread, PackIdenticalLunches, SearingMeat, StirVegetables
- composite-unseen (never used for training or tuning): ArrangeTea, CuttingToolSelection, RecycleBottlesByType, WaffleReheat, WeighIngredients

Training conditions (per model, identical everything except the switches)

id	stage loss	box prompts in training input	location loss
S	on	off	off
BL	off	on	on
F	on	on	on

Shared: same official checkpoint init; frozen encoders/base weights; rank-8 LoRA on the same four late DiT projections plus stage head (16 classes) and location head (3 cameras x 2); identical demonstrations (20 train episodes per seen task = 100, SHA256-ranked, fixed on 2026-11-06), identical annotated window panel, identical sampling order (seed 20261106), microbatch 1 x accumulation 8, AdamW lr 1e-4 wd 0.01, identical optimizer update count within a model (set from a timing calibration *before* any run; the same count is then used for S, BL and F of that model). Loss = action flow-matching MSE (+ Awomo future-latent MSE) + 0.1 stage CE (if on) + 0.1 location smooth-L1 (if on). No stage text is added to the policy instruction in any condition. The box is a 2-px red rectangle drawn on the current frame only; history and future-prediction targets stay clean. No action-only retrained baseline (user decision).

Annotations Automatic pseudo-labels: Grounding DINO (base) proposals → SAM 2.1 (hiera-small) tight mask boxes → Qwen3-VL-4B-Instruct set-of-mark selection of the described target instance per camera (or "not visible"). No human review. Windows whose target is not visible in any camera (and the 'task complete' stage) are excluded from the panel for ALL arms. A random audit sample of the final labels is inspected visually and its hit rate reported.

Evaluation (primary)

- Ordinary inputs for every condition: original task instruction, clean RGB, state. No feedback of predicted heads.
- Conditions evaluated per model: official checkpoint (reference), S, BL, F → 12 policies.
- 10 tasks x 20 fixed seeds (0-19) per policy = 200 episodes per policy, 2,400 total. Same seed ⇒ same initial scene (checked by first-frame hash); official per-task horizon and official success function; policy query every 16 env steps (native chunking); policy RNG seed 4242 + replan index.

- Primary endpoints (per model, paired by (task, seed), pooled over the 10 tasks): (a) F - BL : contribution of stage supervision given box+location, (b) F - S : contribution of box+location given stage. Secondary: F - official (total short-budget adaptation, confounded with extra action training), S - official, BL - official; seen vs unseen subsets; per-task tables.
- Statistics: paired success-rate difference with 95% CI from a stratified (by task) paired bootstrap (10,000 draws), exact two-sided McNemar/sign test on discordant pairs, Holm correction over the two primary contrasts per model. A contrast counts as "supported" only if the Holm-corrected interval excludes 0 and the point estimate is $\geq$ +5 pp.
- All episodes are reported (success / failure / error / timeout). No task, seed or checkpoint is replaced after seeing results. Only the final checkpoint of each run is evaluated.

Optional secondary study (only if time permits, reported separately) Same F checkpoint with self-generated stage/box feedback vs ordinary inputs; not part of the primary claims.

Compute window Training and evaluation on hkust-cluster GPUs 0,1,4,5,6,7 (user authorized exclusive use on 2026-11-08). [One administrative scheduling line is omitted in this public version.]

B. Target definitions

Every stage description in the five training tasks (198 task-specific descriptions over the training and development demonstrations; 195 distinct strings, since "navigate to the stove" occurs in two tasks and "task complete" in three) is mapped to a detector class, an instance hint for the selector, and a role by the rules in Table S1; the three "task complete" stages have no target and their windows are excluded for all arms. No description was left unresolved. Grounding DINO [2] (base) runs with box and text thresholds 0.2, NMS 0.65 and at most six proposals per camera; SAM 2.1 [3] (hiera-small) refines each proposal with a box prompt and the tight box of the mask replaces the proposal. Qwen3-VL-4B-Instruct [1] sees the three cameras upscaled to 512×512 with numbered coloured boxes (set-of-mark prompting [4]), the task instruction, the stage text and the instance hint, and returns one candidate number (or -1) per camera with greedy decoding. Windows: of the 8,931 queued training windows, 8,174 have a confirmed target in at least one camera and form the shared panel; 757 were excluded. Across training and development demonstrations together, 841 windows had no visible target, 606 of them navigation stages whose target was out of view.

C. Label audit

A simple random sample of 48 panel windows (seed 7: 14 navigate, 11 pick, 10 place, 10 execute, 3 wait and 0 press windows) was rendered with the selected box in each camera, eight windows per review sheet, and inspected visually by the AI agent that executed the study (Claude Fable 5.1) after labelling and before any training result was known. The 48 windows contain 103 camera views with a visible-target label.

- 93 of 103 boxes (90%) bound the target object. Eight of these bound the whole appliance rather than the named part (the toaster instead of its lever, the stove top instead of one burner); their centre lies on the correct object but not on the named part, so 85 of 103 boxes (83%) bound the named part itself.
- 10 boxes are wrong: six bound a robot link or the gripper (typically when the arm occludes the target), three bound the wrong instance among duplicates (e.g. a container holding fish instead of the empty one), one bounds unrelated furniture.
- Views labelled "not visible" were not audited for misses; navigation stages have the highest miss rate because the target is often outside the field of view, which is why 606 of the 841 windows without a visible target (training and development demonstrations together) are navigation windows.

These rates characterise the supervision the BL and F arms actually received: the location loss is trained on the target object in roughly nine views in ten and on the named part in roughly eight in ten. Figure 2 of the main paper shows six labelled windows, including typical errors. The six review sheets and the list of sampled windows will be released with the code; the per-window audit decisions were not recorded separately.

D. Training details and curves

Adapter placement. For XR-1 the adapters sit on the last four `attn.qkv_proj`/`attn.o_proj` projections of the action DiT, for Awomo-0.5I on the last four `to_q`/`to_v`/`to_qkv_mlp_proj` projections of its image-action DiT, for Awomo-0.5V on the last four `self_attn.q`/`self_attn.v` projections of its video-action DiT. Rank 8, A initialised Kaiming-uniform, B zero, so every arm starts as the exact official policy (verified by comparing zero-update exports with recorded original actions). The stage head is a $d\times16$ linear layer and the location head a $d\times6$ linear layer followed by a sigmoid, both on the hooked feature.

Initialisation of the new parameters. For both Awomo models the two heads start from the same random draw in all three arms: at update 0, BL and F (identical inputs) have identical action and location losses, and S and F differ in stage loss by at most 0.03 nats, which F's boxed frame explains. For XR-1 they do not: BL and F have identical action losses but location losses up to 0.065 apart, and S and F differ in stage loss by up to 3.9 nats; by the same seeding path the LoRA A matrices of XR-1 may differ between

Stage kind	Pattern	Detector class	Instance hint / role
pick	"pick up the X from the Y"	X (pan→frying pan, bread→slice of bread)	the X about to be picked up / manipulated object
place	"place the X in/on the Y"	Y: tupperware→plastic food container, burner→stove burner, pan→frying pan, pot→cooking pot, plate, glass cup	the destination as written (e.g. "the empty tupperware", "the front-left burner") / destination
execute	"open the drawer . . . "	drawer handle	interaction part
execute	"turn on the P burner"	stove knob	the knob controlling the P burner / interaction part
execute	"stir the vegetables in the pot"	cooking pot	interaction part
press / wait	"push down the lever", "wait for the toaster . . . "	toaster lever	interaction part / watch target
navigate	"navigate to the X"	cup→glass cup, plate, fridge→refrigerator, tupperware→plastic food container, stove, counter→kitchen counter	X as written / navigation target
done	"task complete"	—	excluded

Table S1. **Target rules.** Rules that map a stage description to the detector class and the instance hint used by the selector.

arms as well. Because B starts at zero, the action pathway of every arm still starts identical to the official policy, but XR-1's between-arm differences also contain seed-to-seed variation of the new parameters (main paper, Section 6).

Window construction. A window at anchor t (multiple of 16, $t+32$ inside the episode) contains the history frames each model natively consumes (XR-1: $t{-}6, t{-}4, t{-}2, t$; Awomo-0.5I: every 40th frame back to $t{-}360$ plus the frame $t{+}16$ as the future image target; Awomo-0.5V: every 4th frame from $t{-}96$ to $t{+}32$, the future part being the video target), the corresponding states, the 32 future actions in the raw LeRobot layout, the official `subtask_idx` at t and, for BL and F, the pseudo-label box drawn on the three current frames only. Actions are converted to each model's native layout and normalisation exactly as in its official evaluation code; Awomo-V targets are resampled from 20 Hz to 15 Hz with nearest-neighbour gripper and mode channels, as in its release.

Optimisation and coverage. AdamW, learning rate 10^{-4}, weight decay 0.01, gradient clipping as configured by LeRobot's default preset, one window per micro-batch, eight micro-batches per update, 1,000 updates, data order fixed by the seed 20261106 and identical across arms and models. Micro-batches cycle through the five tasks and draw a window of that task with replacement, so a run makes 8,000 task-balanced draws (1,600 per task) that reach 4,977 distinct windows, 60.9% of the 8,174-window panel (DeliverStraw 77%, GetToastedBread 62%, PackIdenticalLunches 58%, SearingMeat 50%, StirVegetables 69%); it is not one pass over the panel. Of the 800 draws in the last 100 updates, 296 (37%) were first visits of their window. The flow-matching time $t \sim \mathcal{U}(0,1)$ and the noise are drawn from a seed that depends only on the micro-batch index, so S, BL and F see identical noise. Wall-clock per run on one H100 shared with evaluation workers: 40–41 minutes for XR-1, 106–173 minutes for the Awomo models. Figure S1 shows the loss terms.

E. Additional results and episode-level data

Per-task counts and splits. Table S2 lists every per-task success count behind Figures 3 and 4 of the main paper. Table S3 splits each paired contrast into the five composite-seen tasks, whose demonstrations were used for adaptation, and the five composite-unseen tasks. Table S4 gives episode statistics and Table S5 the rescue/regression counts of the full arm against the official checkpoint.

Scene identity. For 191 of the 200 seeds the first observation (SHA-256 of the left agent-view frame) and the initial state are identical across all twelve primary policies. For the other 9 seeds (SearingMeat 1, 4, 6, 11, 14; StirVegetables 1, 11, 14, 16) the first frame came in two variants across policies, and for 2 of them (StirVegetables 1 and 11) the initial robot base pose also differed. The pre-registered analysis keeps all 200 pairs; Table S6 repeats every contrast, post hoc, on the 191 identical seeds. No contrast becomes supported, and every contrast keeps its direction and its significance at the 0.05 level with two exceptions: for Awomo-0.5V, F$-$S also passes the Holm step ($p = 0.031$, against 0.060 on all 200 seeds), and Awomo-0.5I's BL$-$official interval stays below zero ($[-18.3, -0.5]$) while its McNemar p rises to 0.054 (0.033 on all 200 seeds).

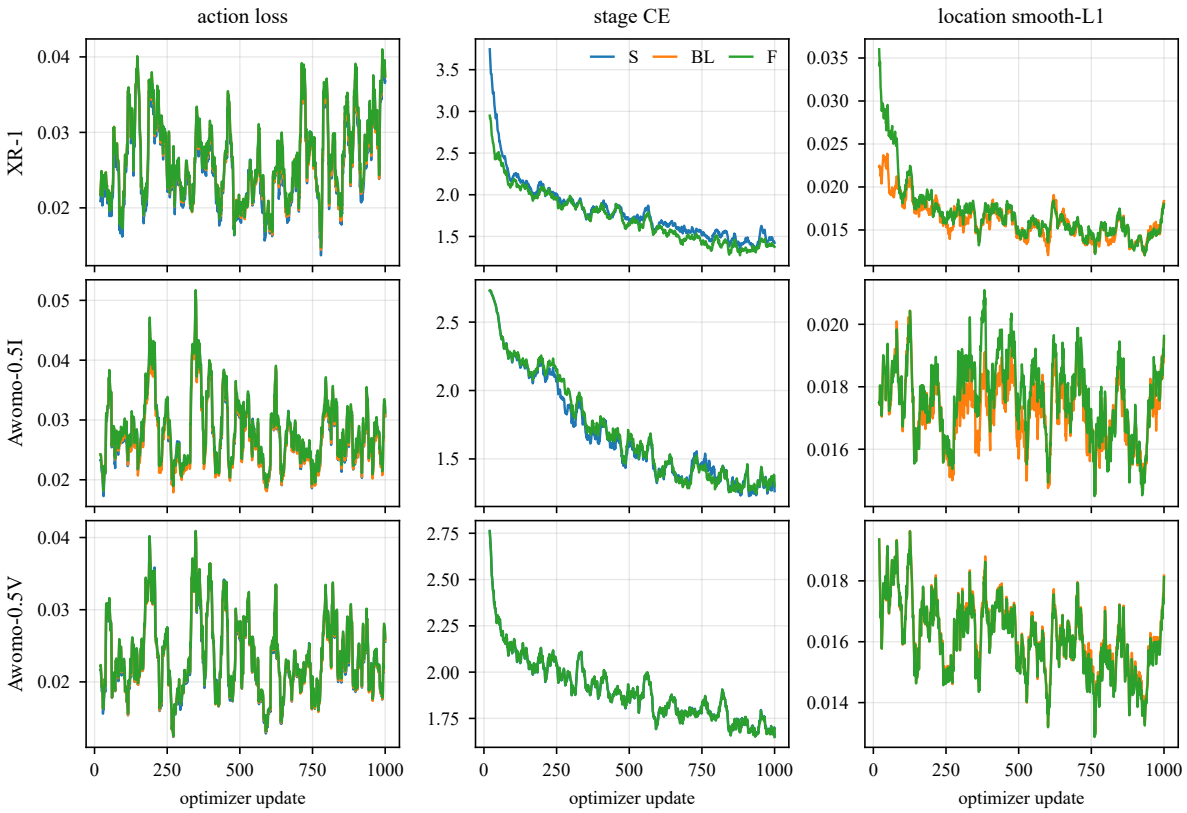


Figure S1. **Training-loss terms.** Per model (rows) and arm, 20-update running mean. Stage cross-entropy and location smooth-L1 are only shown for the arms in which they are switched on. F is drawn last, so where the arms' curves coincide (most of the action-loss panels, Awomo-0.5V's stage cross-entropy) S and BL are hidden under it.

		composite-seen					composite-unseen							
Model	Policy	Straw	Toast	Lunch	Sear	Stir	Tea	Cut	Recycle	Waffle	Weigh	Seen	Unseen	All
XR-1	official	10	2	5	5	6	10	9	1	12	8	28	40	**34.0**
	S	7	5	8	9	8	14	12	0	12	6	37	44	**40.5**
	BL	7	6	8	7	9	9	13	3	10	9	37	44	**40.5**
	F	10	6	8	9	8	9	11	2	12	8	41	42	**41.5**
Awomo-0.5I	official	12	6	5	4	10	10	12	6	12	12	37	52	**44.5**
	S	13	6	5	8	6	11	9	2	10	9	38	41	**39.5**
	BL	14	4	9	4	4	8	11	3	6	6	35	34	**34.5**
	F	10	4	9	6	8	7	9	4	9	10	37	39	**38.0**
Awomo-0.5V	official	13	10	7	6	8	13	7	9	7	8	44	44	**44.0**
	S	9	7	10	4	12	11	6	8	9	7	42	41	**41.5**
	BL	8	5	8	4	9	12	3	8	8	8	34	39	**36.5**
	F	10	6	1	4	9	8	5	4	9	10	30	36	**33.0**

Table S2. **Per-task success counts.** Successes out of 20 seeded episodes per task, with seen, unseen and overall success rates (%) for the official checkpoint and the three adapted arms of each model. All arms are evaluated with ordinary inputs on the same 200 seeded episodes.

Episode-level data. Each of the 2,400 episodes is logged as one JSON record with model, condition, task, seed, split, instruction, horizon, success flag, step count, number of policy queries, median query latency, wall-clock time, and the SHA-256 of the first observation and the initial state (used for the scene-identity check above); the first and last agent-view frames and a 5 fps agent-view video are stored alongside each record.

Model	Contrast	seen tasks Δ (pp)	seen tasks 95% CI	unseen tasks Δ (pp)	unseen tasks 95% CI
XR-1	F−BL	+4	[−7, +16]	−2	[−11, +7]
	F−S	+4	[−6, +14]	−2	[−10, +6]
	F−official	+13	[+2, +24]	+2	[−6, +10]
	S−official	+9	[−2, +20]	+4	[−4, +12]
	BL−official	+9	[−2, +20]	+4	[−3, +12]
Awomo-0.5I	F−BL	+2	[−8, +13]	+5	[−5, +15]
	F−S	−1	[−13, +11]	−2	[−12, +8]
	F−official	0	[−11, +11]	−13	[−23, −3]
	S−official	+1	[−11, +13]	−11	[−22, 0]
	BL−official	−2	[−15, +10]	−18	[−29, −7]
Awomo-0.5V	F−BL	−4	[−15, +7]	−3	[−13, +7]
	F−S	−12	[−21, −3]	−5	[−15, +5]
	F−official	−14	[−25, −3]	−8	[−20, +4]
	S−official	−2	[−14, +10]	−3	[−13, +7]
	BL−official	−10	[−21, +1]	−5	[−16, +5]

Table S3. **Contrasts split by task group.** Paired success differences on the five composite-seen tasks, whose demonstrations were used for adaptation, and on the five composite-unseen tasks (100 pairs each), with task-stratified paired-bootstrap 95% intervals (10,000 draws).

Model	Policy	steps[a]	timeouts	time (s)[b]
XR-1	official	1,365	132	200
	S	1,427	119	226
	BL	1,332	119	206
	F	1,281	117	228
Awomo-0.5I	official	1,314	111	326
	S	1,351	121	322
	BL	1,585	131	353
	F	1,387	124	328
Awomo-0.5V	official	1,372.5	112	385
	S	1,422	117	380
	BL	1,319	127	394
	F	1,271	134	376

Table S4. **Episode statistics.** [a]Median length (steps) of the successful episodes. [b]Mean wall-clock per episode, including simulation and policy inference on shared GPUs. Every failure is a horizon timeout (no early terminations occurred).

Model	F rescues	F regressions	net
XR-1	33	18	+15
Awomo-0.5I	24	37	−13
Awomo-0.5V	27	49	−22

Table S5. **Rescues and regressions.** Episodes the F arm solves that the official checkpoint fails (rescues) and vice versa (regressions), out of 200 paired episodes.

Crashed and unreleased episodes. Eighteen Awomo-0.5V official-checkpoint episodes crashed with a GPU out-of-memory error. Their error rows were removed and the episodes rerun from scratch, a decision taken after partial results had been viewed; the pre-registration instead reports errors as such. The original shard files, including the 18 error rows, are released with the logs; the 132 completed rows in them are unchanged in the released results, and 5 of the reruns succeeded. Counting the crashes as failures gives 83 instead of 88 official successes, and S−, BL− and F−official of 0.0 $[-8.0, +8.0]$, -5.0 $[-13.0, +2.5]$ and -8.5 $[-17.0, -0.5]$ pp (exact McNemar $p = 1.000$, 0.275 and 0.068; task-stratified bootstrap, 10,000 draws); the primary contrasts do not involve this checkpoint. Nine further official-checkpoint episodes are not among the 2,400: one smoke-test episode (XR-1, DeliverStraw, seed 0) and eight from a first launch that was split by task group and was aborted and relaunched with the final shard split after its first outcomes had been seen. All nine have the same outcome, step count and first-frame hash as the released episodes of the same (model, task, seed), and their rows are released too. No episode was dropped.

The analysis scripts recompute every number in the paper from these records, the training logs and the label files, except the label-audit counts (Section C), the number of discarded feedback episodes (73), the number of queued windows and the trainable-parameter counts, which come from the cluster run logs.

Model	Contrast	Δ (pp)	95% CI	p	Holm p
XR-1	**F−BL**	+2.1	[−5.2, +9.4]	0.678	1.000
	F−S	+2.1	[−4.7, +8.9]	0.659	1.000
	F−official	+7.9	[+1.0, +14.7]	0.044	
	S−official	+5.8	[−1.0, +12.6]	0.152	
	BL−official	+5.8	[−1.0, +12.6]	0.144	
	S−BL	0.0	[−6.3, +6.3]	1.000	
Awomo-0.5I	**F−BL**	+3.1	[−4.2, +11.0]	0.504	1.000
	F−S	−1.0	[−9.4, +7.3]	0.902	1.000
	F−official	−6.3	[−14.1, +1.6]	0.155	
	S−official	−5.2	[−13.6, +3.1]	0.295	
	BL−official	−9.4	[−18.3, −0.5]	0.054	
	S−BL	+4.2	[−2.6, +11.5]	0.322	
Awomo-0.5V	**F−BL**	−3.1	[−11.0, +4.7]	0.526	0.526
	F−S	−9.4	[−16.2, −2.6]	0.015	0.031
	F−official	−11.0	[−19.4, −2.6]	0.019	
	S−official	−1.6	[−9.4, +6.3]	0.801	
	BL−official	−7.9	[−15.7, 0.0]	0.077	
	S−BL	+6.3	[−2.1, +14.7]	0.182	

Table S6. **Sensitivity to scene identity (post hoc).** Every contrast recomputed on the 191 of 200 seeds whose first observation and initial state are identical across all 12 policies (task-stratified paired bootstrap, 10,000 draws; exact McNemar p, uncorrected). Bold: pre-registered primary contrasts, with the Holm-corrected p; none of them is supported under the pre-registered rule.

Feedback study. The secondary feedback evaluation (Table 3 of the main paper) is logged in the same format, one shard per model and GPU, and analysed by `analysis/feedback.py`, which pairs each FB episode with the ordinary-input F and official episodes of the same (task, seed) and writes the table and the numbers quoted in the text. It draws only the location-head box; the stage head is not used, although the pre-registration mentions stage/box feedback. All three runs are complete (200 episodes each). Some FB episodes started from the other first-frame variant than the F episode of the same seed (4, 6 and 5 for XR-1, Awomo-0.5I and Awomo-0.5V); without them FB$-$F is -10.2, -5.2 and -4.6 pp ($p = 0.012$, 0.253 and 0.314; 196, 194 and 195 pairs). An earlier Awomo-0.5V run (73 episodes) read the location head from the wrong feature and was discarded without analysis.

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